Overlap and countability

Johan Rooryck Camil Staps

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Abstract

In English, some classifier constructions can be modified by numerals ((*two*) *slices of cake*), while others cannot be so modified outside of contexts of coercion ((#*two*) *hints of cognac*). Countability is also excluded in phrases like *the* (#*two*) *outlines of a house* or (**two*) *feelings of helplessness*. We propose that these phrases have two possible counting bases: in one, the portions or the kind of the complement (*cake*, *cognac*, *a house*, *helplessness*) are counted (as in Chierchia's 2010 analysis of *quantity* (*of water*)); in the other, the classifiers themselves (*slices*, *hints*, *outlines*, *feelings*) are counted. We argue that different types of overlap in the elements of the counting base can prevent countability, in either the low or the high reading, or in both. For *slice*, both the high and the low reading provide a suitable counting base; for *hint* and *outline*, only the high reading provides a suitable counting base; for *feeling*, neither does. This analysis suggests that overlap is an important ontological property in refining the mass/count distinction.

Keywords: overlap, classifiers, numerals, countability, mass/count

1 Introduction

Nouns are traditionally categorized into MASS and COUNT. In English, count nouns have a plural, can take the indefinite article, and can be modified by numerals (*a cat*, *two cats*), but mass nouns do not (**a blood*, **two bloods*). Some nouns, however, display conflicting behavior and fall in between.

In this article, we look specifically at the countability properties of English classifier expressions (in the sense of Lyons 1977[1]), which are constructions of the form *Cl of N*, such as *drop of water*.[2] The classifier (here *drop*) is typically a count noun and the complement (here *water*) is typically a mass noun. In the case of *drop of water*, the entire expression is count, as is also the case with the following classifiers:[3]

(1) a. *a drop/splash* / (*three*) *drops/splashes of water*

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[1] We agree with Aikhenvald (2025: pp. 119–120) that the English nouns we look at here differ from "true" numeral classifiers in terms of syntactic distribution. However, our primary focus is semantic, and on this plane the meanings of the nouns under discussion are highly similar to "true" classifiers as found in many languages across Asia, Oceania, and the Americas: both "true" classifiers and English classifiers select for / contribute properties of shape and dimensionality, as already observed by Lyons (1977). The classifiers we discuss in this article, with representative examples in (1)–(4), are united by the fact that the classifier tells us something about the way the complement presents itself in its environment.

[2] Keizer & Ten Wolde (2024) call phrases like *bird of prey*, *wall of clay*, and *man of honour* "head-classifier constructions". In these constructions, the second noun "classifies" the first. This construction is distinguished from "pseudo-partitives" like *cup of water* where the first noun is a measure and the second noun functions as the head. When we talk about classifiers, we mean the first noun in such "pseudo-partitives", which have a semantic contribution comparable to that of classifiers in "true" classifier languages (see footnote 1).

[3] At least some of these classifiers also have a purely quantificational reading: this reading is active in sentences like *a splash of vinegar amounts to roughly one tablespoon*. We are interested strictly in the non-quantificational reading, where *a splash of vinegar* really takes the shape of a splash.

b. *a slice* / (*four*) *slices of cake*

c. *a lump* / (*two*) *lumps of sugar*

There are also classifiers that display conflicting behavior. They can be modified by the indefinite article and can be pluralized, but cannot be modified by numerals:

(2) a. *a hint* / (#*two*) *hints of cognac*

b. *a tinge* / (#*two*) *tinges of green*

c. *a whiff* / (#*two*) *whiffs of perfume*

Rooryck (2024), from where the above examples have been taken, describes the classifiers in (1) as Samples and those in (2) as Pointers. The rationale behind this terminology is that the expressions in (1) describe portions, i.e. samples, of a material that they are coexistent with; whereas the expressions in (2) describe things that only provide evidence for, i.e. point to, the existence of a material. Thus, the classifiers in (2) typically imply an experiencer (and are therefore to an extent subjective), whereas the classifiers in (1) refer to entities that can be said to exist independently. However, they are similar in that both describe the ways in which a material interacts with its environment: *water* forms *drops* and *splashes* due to its material properties (roughly, being a liquid); *perfume* is experienced in *whiffs* due to its material properties (being a gas). Although most classifiers seem to refer to small quantities, there are examples of both with a size-neutral or large default interpretation (*sliver*/*slice*/*slab of cake*; *whiff*/*waft*/*wave of perfume*).

The Pointers in (2) defy a strict division between mass and count. Like count nouns, they can be modified by the indefinite article and can be pluralized. However, like mass nouns, they cannot be modified by numerals. A closer look at their properties provides useful information about the way humans categorize real world entities.

Informally, we propose that a *hint of cognac* is an entity that is a "manifestation" of a portion of cognac: it is related to that portion but not coexistent with it (unlike a *drop of water*, which consists of water). The plural *hints of cognac* has a high and a low counting base. With the high counting base, countability depends on properties of the Pointer itself (*hint*). The subject may experience multiple manifestations of cognac which are separated in time. On this reading, hints can be counted: at a dessert tasting one might say, *I tried three desserts, but was only able to taste two hints of cognac* (hence the # judgment in (2) above). In this reading, the hints are manifestations of distinct portions of cognac. With the low counting base, countability depends on properties of the underlying portion of cognac. When a dessert is flavored with *hints of cognac*, the various hints (such as olfactory and gustatory) are manifestations of the same portion of cognac in the dish. We suggest that it is the overlap in reference to the same underlying portion that prevents countability in this case: we cannot describe a single dessert as manifesting **two hints of cognac*.

The class of Pointers extends beyond what is typically seen as classifier constructions. In our view, they also cover certain constructions with a regular DP complement (3) or an abstract noun complement (4). These display similarly problematic countability patterns:

(3) a. *the outline* / (#*two*) *outlines* (*of a house*)

b. *the contour* / (#*two*) *contours* (*of a project*)

(4) a. *a feeling* / (#*two*) *feelings* (*of helplessness*)

b. *a depth* / (#*two*) *depths* (*of meaning*)

c. *a note* / (#*two*) *notes* (*of peach*)

For (3a), for instance, the default interpretation of plural *the outlines of a house* is that of a detailed floor plan showing the outlines of each individual room. This contrasts with the interpretation of the singular *the outline of a house*, which can be represented by as little as two vertical lines for the walls, a horizontal line for the base, and two mirrored slanted lines for the roof.[4] This reading does not allow modification with a numeral. However, modification by numerals is possible in certain contexts, which correspond to the high counting base reading, for instance when drawing two 5-line outlines of the same house. For the expressions in (4), neither the low nor the high reading allows for modification by numerals. We argue that all these facts can be derived from the presence or absence of different types of overlap in the various counting bases.

The rest of the paper is structured as follows: after covering the relevant background on mass and count in section 2, we first provide our analysis of Samples of material in section 3. The following sections explain the countability patterns observed with Pointers to material (section 4), classifiers with DP complements (section 5), and classifiers with abstract complements (section 6). Section 7 provides a comparison with Metaphorical Measure Expressions (Sutton et al. 2022), which display similar countability patterns but should be kept apart on syntactic grounds. Section 8 takes us out of the realm of classifier expressions to show that the notions we introduce here have wider applicability. Section 9 concludes.

2 Background on mass and count

In common approaches to pluralization (Chierchia 2010, a.o.), plurals denote sets of entities. The ability to pluralize and count depends on whether these sets consist of atomic individuals. Thus, *two cats* denotes sets of two individuals (for instance, *Yosha* and *Kaz*), which cannot be further subdivided into cats (there is no proper subpart of *Yosha* that is also a cat). On the other hand, *water* denotes a set of portions of water. This set does not have stable atoms: in some contexts, a drop of water is not enough to constitute water (*there is no water in the tank*), but in other contexts, a drop of water does constitute water (*add a drop of water to the mix*). Thus, the terms "mass noun" and "count noun" are abbreviations for ontological properties: "mass" refers to the property "lacks stable atoms", and "count" to the property "has stable atoms".

These ontological properties are reflected in grammar. In English, typical count nouns can be modified by the indefinite article, be pluralized, and modified by numerals; typical mass nouns cannot. It has been long known that this binary division is too simplistic (e.g. Allan 1980). Indeed, nouns like *hint*, *feeling*, and *contour* are problematic: they combine with the indefinite article and can be pluralized, which suggests that there are stable atoms; on the other hand, they cannot be counted, which suggests that there are none. Thus, our three syntactic tests do not always align, suggesting that the ontology behind the mass/count distinction is more complicated than the one sketched above. In offering an explanation why nouns like *hint*, *feeling*, and *contour* may not combine with numerals (in spite of being otherwise count) we hope to clarify some of the ontological properties at work.

Before turning to our analysis of these problematic nouns, we will first discuss Samples like (*two*) *drop*(*s*) *of water*, which are unproblematically count. The way we see these Samples is based on Chierchia's (2010: pp. 124–128) treatment of the "relational noun" *quantity* (*of*), which we briefly introduce here.

In Chierchia's (2010) approach, *quantity* is a variable over partitions Π with the following properties:

(5) a. $\Pi(\mathrm{P}) \subseteq \mathrm{P}^{+}$
A partition of P is a total subproperty of P.
b. $\forall x[\Pi(P)(x) \rightarrow \forall y[y < x \rightarrow \neg\Pi(P)(y)]]$
Relative atomicity: if x is a member of a partition of P, no proper part of x is.
c. $\forall x[\Pi(P)(x) \rightarrow \forall y[\Pi(P)(y) \rightarrow \neg\exists z[z \leq x \wedge z \leq y]]]$
No overlap: no two members of a partition overlap.

[4] This visual distinction is confirmed by an online image search for "the outline of a house" vs. "the outlines of a house".

For instance, in a context with five apples (a, b, c, d, and e), *quantity*$_1$ *of apples* may refer to $\Pi_1(\textit{apples}) = [a \cup b, c \cup d \cup e]$.[5] In this case, $a \cup b$ and $c \cup d \cup e$ are stable atoms for *quantity*$_1$ *of apples*. Similarly, if a–e are portions of water, $a \cup b$ and $c \cup d \cup e$ may be stable atoms for *quantity*$_2$ *of water*. This explains why we can talk of *two quantities of water* even though *water* is a mass noun. Neither quantity of water is atomic in an absolute sense, but they are atomic relative to the context-specific partition.

Specifically, distributional properties of mass and count nouns can be formalized as restrictions on modifiers. For example, the indefinite article *a* could be seen as being restricted to atoms (Chierchia 2010: p. 135), using the definition of atomicity in (7) (Chierchia 2010: p. 113):

(6) $[\![a]\!](P) = \lambda Q \exists x[P(x) \wedge Q(x)]$, if $AT(P) = P$ (else undefined)

(7) a. $AT(P) = \{x \in P : \forall y \in P[y \leq x \rightarrow x = y]\}$, for P of type <e, t>
The atoms of a property P are those P-satisfying entities of which no proper subpart also satisfies P.

b. $AT(x) = AT(\lambda y[y \leq x])$, for x of type e
The atoms of an entity x are the atoms of the property of being a subpart of x.

In the universe above, $AT(\{a \cup b, c \cup d \cup e\}) = \{a \cup b, c \cup d \cup e\}$, so *a quantity of water* is grammatical. Chierchia suggests that "a similar analysis can be devised for other quantifiers and for the numerals" (2010: p. 135). Thus, the denotation of a numeral could be similarly restricted that have non-overlapping atoms (cf. Chierchia 2010: p. 124):

(8) a. $[\![two]\!](P) = \lambda Q \exists x[P(x) \wedge Q(x) \wedge \mu_{AT,P}(x) = 2]$

b. For any x, P, $\mu_{AT,P}(x)$ is defined only if
i. P(x) holds;
ii. P has stable atoms; and
iii. the P-atoms in x do not overlap: $\forall a, b \in x \cap P[a \neq b \rightarrow a \cap b = \varnothing]$;[6]
if defined, $\mu_{AT,P}(x)$ = the number of stable P-atoms that are part of x.

Condition (8b-iii) is an addition to Chierchia. We would not want to count $\{a, b, a \cap b\}$ as three quantities, and whereas Chierchia builds this into the definition of partitions (5c), we will see in sections 4 to 6 that it is useful to add this as a condition on $\mu_{AT,P}$ as well. It will also become clear there that the weaker restriction of quantization (that atoms do not stand in a proper part relation to each other) is not enough for the cases discussed here.

3 Samples: *drop*

We first address the unproblematically count classifiers like *drop*, *slice*, and *lump* (Samples in Rooryck's 2024 terms). On the one hand, these are relational nouns like *quantity* and therefore likely to involve a partition of their complement. On the other hand, at least some of these classifiers can appear without a complement: *there are lumps in the batter*. Therefore, these classifiers have lexical semantics that cannot be specified in terms of their complement. These lexical semantics typically involve "material and mechanical properties of consistency (e.g., plasticity, density, viscosity, texture, or cohesion of matter)" (Rooryck 2024: p. 6). This is a difference between Samples and Chierchia's running example *quantity*:

[5] This partition has the properties in (5): it is total (each of a–e are a member of an element of the partition); it has relative atomicity (no subpart of the atoms $a \cup b$ and $c \cup d \cup e$ are in the partition themselves); and there is no overlap between the elements of the partition ($a \cup b$ and $c \cup d \cup e$ are disjoint).

[6] Cf. Grimm's (2012) notion of overlap.

the latter is a "light classifier" in the sense that it refers to a portion of material but does not provide any information about its physical properties.[7]

To account for the lexical semantics of Samples, we propose that these classifiers straightforwardly denote sets of entities with the appropriate physical properties. For instance, *drop* denotes the set of entities that have the physical properties of a drop. These properties are such that they require a portion of material which stands in a CONSISTS-OF relationship to the classifier entity. Let $S_{Cl,M}$ stand for the set of Samples of classifier Cl and material M. We then define:

(9) $\text{material}(x \in S_{Cl,M}) =_{def} \bigcup\{p \in M \mid p \text{ is contained in } x\}$

The classifier entity ($x \in S_{Cl,M}$) is a count entity; the material it consists of (a subset of M) is a portion of a mass entity. Thus, the consists-of relation can be seen as a link between the domains of mass and count. We assume this relation is the contribution of the highly polysemous preposition *of*.

Consider an example world in which $[\![\textit{water}]\!] = \{a, b, c, d, e\}$ and $[\![\textit{drop}]\!] = \{p, q, r, s, t\}$, p consists of $a \cup b$, and q consists of c. The other drops (r, s, and t) are drops that consist of portions of other materials; the other portions of water (d and e) do not form drops. In this universe, the denotation of *drop of water* will be $S_{drop,water} = \{p, q\}$.

For a consists-of relationship it is natural to require that Samples consist of non-empty portions, and that the portions of distinct Samples do not overlap.

(10) Samples consist of non-empty portions:
$\forall x \in S_{Cl,M}[\text{material}(x) \neq \varnothing]$

(11) The portions that distinct Samples consist of do not overlap:
$\forall x, y \in S_{Cl,M}[x \neq y \rightarrow \text{material}(x) \cap \text{material}(y) = \varnothing]$

From this follows the fact that the portions of a set of Samples of the same material form a subpartition over that material:

(12) $\forall Cl, M \exists \Pi[\{\text{material}(x) \mid x \in S_{Cl,M}\} \subseteq \Pi(M) \wedge \Pi \text{ is a partition over } M]$

There are now two ways to approach the countability properties of Sample phrases: the high and the low reading. They differ in the noun that contributes countability properties: the noun for the material in the low reading, or the classifier in the high reading. For *drop of water*, both the high and the low reading give rise to a straightforwardly count expression.

The LOW COUNTING BASE reading is used when we use a phrase like *two drops of water* to refer to $\{p, q\}$ from the example above. In this case, the fact that *drop of water* is countable can be explained in the same way as the countability of *quantity of water*: each drop refers to a distinct portion, which serves as a stable atom (i.e., the stable atoms are $a \cup b$ and c). This is the low counting base reading because it depends on the properties of the material in the complement.

On the other hand, we might say that *drop of water* is countable because *drop* itself is. In *I heard two drops on the roof*, there is no complement, and so the countability of *drops* must derive from the fact that the drops are separated in time (rather than space). It seems reasonable to say, then, that the countability of *drops* in *I heard two drops of water on the roof* does not depend on the division of water into portions but, again, on the temporal separation of the drops.[8] This is the HIGH COUNTING BASE reading.

In the next section we will see that the two counting bases lead to different countability behavior for classifiers like *hint*. It can be hard to tease apart the high and the low reading for Samples like *drop*, because for these classifiers a temporal division (characteristic for the high reading) usually corresponds

[7] In the same way that a light noun (*thing*) does not provide any more information beyond noun-ness and a light verb (*do*) does not provide any more information beyond verb-ness.

[8] Chierchia (2010) does not distinguish between the high and the low reading. His analysis corresponds to the low reading.

to a division in material (characteristic for the low reading): a *drop* only falls once. However, there are cases where a Sample is clearly used with a high reading. In the context of a baptism, a priest may pour *three scoops of water* over the head of the person to be baptized. Typically, the water of the first scoop falls back into the basin before the second scoop is taken, and so the portions of the three scoops are not necessarily fully disjoint. Furthermore, *three scoops* is not a simple measure (in the way *two pounds* can be equal to *one kilo*): it is relevant that the three instances are separated, since they symbolize the three parts of the Holy Trinity. Thus, here we have a case where it is clearly not partitioning of the portions (the low reading) but temporal disjointness (the high reading) that gives rise to countability of a Sample phrase.

4 Pointers: *hint*

As pointed out in the introduction, there is also a class of classifiers (called Pointers in Rooryck 2024) which is not straightforwardly countable. These classifiers can be modified by the indefinite article and can be pluralized, but cannot be modified by numerals, as shown in (2), repeated here:

(2) a. *a hint* / (#two) hints of cognac
b. *a tinge* / (#two) tinges of green
c. *a whiff* / (#two) whiffs of perfume

Our analysis of Pointers is based on that of Samples, with one important difference. Whereas Samples like *drop* are physical entities and therefore consist of a portion of a material, Pointers are not: they only *provide evidence for* a portion of a material. We will say that they are MANIFESTATIONS of a portion: a *hint of cognac* is a manifestation of a portion of cognac. The crucial difference is that while two Samples cannot consist of the same portion, two Pointers can be manifestations of the same portion: as pointed out in the introduction, when a dessert is flavored with *hints of cognac*, the various hints (such as olfactory and gustatory) are manifestations of the same portion of cognac in the dish. Thus, the equivalent of (11) does not hold for manifestation (we use $\mathrm{P_{Cl,M}}$ for the set of Pointers of classifier Cl and material M):

(13) $\mathrm{material}(x \in \mathrm{P_{Cl,M}}) =_{\mathrm{def}} \bigcup\{p \in M \mid p \text{ is } \textit{manifested by } x\}$

(14) The portions that distinct Pointers consist of may overlap (contrast (11)):
$\nvdash \forall x, y \in \mathrm{P_{Cl,M}}[x \neq y \rightarrow \mathrm{material}(x) \cap \mathrm{material}(y) = \varnothing]$

As a result, we may imagine a universe where $[\![\textit{cognac}]\!] = \{a, b, c\}$ and $[\![\textit{hint}]\!] = \{p, q\}$, where p is an olfactory and q a gustatory manifestation of the same portion $a \cup b$. In such a case, the two counting bases have a distinct effect. In the high counting base reading, we are counting based on the stable atoms p and q. This counting base is used when making an explicit distinction between two different hints, separated in time, place of experience, or other salient properties of the hints themselves rather than the underlying portion:

(15) *I detected two hints of cognac while tasting this dish: a fruity and a floral note.*

In the low counting base reading, the counting base may contain the same portion twice, namely as the portion that is manifested by two distinct hints. In the example above, $a \cup b$ is manifested by both p and q, so the counting base is $\{a \cup b, a \cup b\}$.[9] Keeping in mind the no-overlap restriction on the use of numerals in (8b-iii), this explains why sentences with a low reading, such as (16), are infelicitous in default contexts:

[9] We assume that the counting base is a multiset, i.e. a set that can contain elements multiple times.

(16) *There are* (**two*) *hints of cognac in this dish.*

The low counting base reading can be salvaged when the context is informative enough to guarantee that the portions manifested by the individual Pointers do not overlap:

(17) *I tried three desserts, but detected only two hints of cognac.* (i.e., one in dessert a and one in dessert b)

4.1 Samples and Pointers: interim summary

In sum, both Samples and Pointers have bipartite denotations: they denote a count entity contributing spatial and/or experiential properties, and a portion of material. For Samples, the count entity consists of the material; for Pointers, it only manifests the material. Because of the spatial dimension present in Samples but not Pointers, the consists-of relation is more restrictive: the portions that distinct Samples consist of cannot overlap.

There are two distinct counting bases that may be used when modifying a classifier phrase. The high counting base reading targets the count entities, focusing on their temporal or cognitive separation: the consecutive scoops of water in a baptism, or complementary sensations experienced when tasting a dish (15). Since this counting base consists of count entities, it always allows for modification by numerals.

The low counting base reading targets the underlying material, focusing on spatial separation. It results in countability for Samples, for which the consists-of relation partitions the material they consist of. However, because Pointers do not partition material – two Pointers may be manifestations of the same, or overlapping, portions – these classifiers do not combine with numerals in this reading.

It seems that the low counting base reading is the default: to get the high counting base reading with Pointers it is necessary to emphasize the temporal or cognitive separation (cf. (15) above). This underscores the primacy of the spatial faculty in the human mind (see e.g. Winter et al. 2015).

5 Pointers with DP complements

We now turn to the analysis of a set of classifier constructions with DP complements, exemplified in (3), repeated here:

(3) a. *the outline* / (*#two*) *outlines* (*of a house*)
b. *the contour* / (*#two*) *contours* (*of a project*)

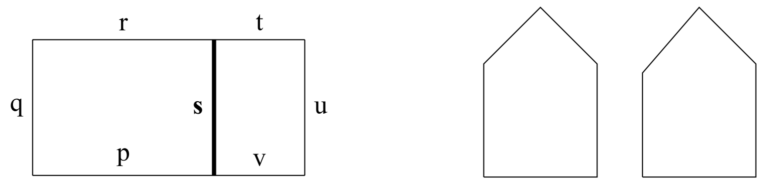
Although it may seem somewhat unnatural to talk about these expressions as "classifier constructions", we see them as a subtype of Pointers: *an outline of a house* is a manifestation of a house, providing evidence for the existence (at least in the drawer's mind) of a house.

As pointed out above, *an outline of a house* refers to a 4 or 5-line sketch, whereas *the outlines of a house* refers to a detailed floor plan. Intuitively, *the outlines of a house* refers to the collection of outlines of the individual rooms of the house. It is possible to draw *three outlines of a house* in the sense of three 5-line outlines, but *three outlines of a house* cannot refer to a floor plan of a house with three rooms.

It therefore seems that the plurality of *outlines* is mapped onto the parts of the complement. Thus, this set of classifiers expressions is naturally quite constrained. First, it requires a classifier like *outline*, where multiple outlines combined form a new outline (or, conversely, where a single outline can be made more specific using sub-outlines). And second, it requires a complement like *house*, for which there is a salient subdivision into parts (rooms) which can also have outlines.

Since we see the expressions in (3) as Pointers, we assume that *outline* has a count denotation which stands in a manifestation relation to the thing it outlines. For instance, imagine a house h with two rooms, a and b. Typically, the rooms share walls: the walls of a are $\{p, q, r, s\}$; the walls of b are $\{s, t, u, v\}$ (see fig. 1a). In this case, the meaning of *outlines of a house* involves two outlines, one a manifestation of $\{p, q, r, s\}$; the other of $\{s, t, u, v\}$.[10] The overlap between these sets violates condition (8b-iii) on the use of numerals. This explains why modification by numerals is excluded in the low reading.

In the high reading, we rely on a separation of the outlines, rather than a separation of the walls they manifest. This corresponds to fig. 1b. In this reading, *outlines of a house* is countable, because it uses the set of outlines (which are count entities) as the counting base.



(a) Low reading: *the* (#*two*) *outlines of a house* (b) High reading: *two outlines of a house*

Figure 1: The low and high reading of *outlines of a house*.

The *contours of a project* is slightly more abstract but works in the same way, dividing up the *project* into subprojects in the same way that a *house* is divided up into rooms. Since the subprojects relate to each other, the contours of the various subprojects will overlap in a similar way as the outlines of the rooms overlap in fig. 1a. This precludes countability in the low reading. To get the high reading, we may imagine convoluted contexts with different kinds of contours, such as a financial contour (the budget) and a temporal contour (the planning). Someone may then have deadlines for two contours of the same project.

6 Pointers with abstract complements

Finally, we turn to classifiers with an abstract noun as a complement, as in (4), repeated here:

(4) a. *a feeling* / (#*two*) *feelings* (*of helplessness*)
b. *a depth* / (#*two*) *depths* (*of meaning*)
c. *a note* / (#*two*) *notes* (*of peach*)

The distinction between Pointers with concrete complements (section 4) and those with abstract complements can be hard to draw. The reason for treating *note*(*s*) *of peach* here (as opposed to *hint*(*s*) *of cognac*) is the fact that a note, even when the complement can be concrete, does not imply the existence of a concrete portion: when a wine is described as having *notes of peach*, this does not mean that peaches were used as an ingredient. Rather, the brewing process has led to the formation of aromatic compounds similar to those found in peaches. We say that the wine manifests the CONCEPT of peach without providing evidence for an actual portion of peach.[11]

Still, even though these expressions refer to concepts rather than portions, we treat them as denoting manifestations, and therefore as Pointers. However, there is a difference with the Pointers discussed in sections 4 and 5, in that modification by numerals seems to be entirely unacceptable. Even when someone

[10] A precise denotation would distract from our main argument, as it would also have to cover the way in which the outlines of the individual rooms are combined into a floor plan. In this way, *outline* is different from homogeneous nouns like *line*, *twig*, or fence (cf. Mittwoch 1988, Krifka 1992, Rothstein 1999, 2010): they denote entities that can also be subdivided into parts that are *lines*, *twigs*, or *fences*, but *outlines* provides additional information which, say, *fences* does not.

[11] There may be speakers for whom *hint* can be used this way, too, or for whom *note* can imply a concrete portion. Some nouns may be used in multiple ways depending on the context.

was feeling helpless on Monday and Wednesday but not Tuesday, using a temporal separation characteristic for the otherwise permissive high reading, it seems odd to say they experienced **two feelings of helplessness*. We therefore need to explain why neither the low nor the high reading allows for countability.

For the low reading, we assume that the abstract complement is kind-denoting (see e.g. Carlson 1980 and Moltmann 2013). This immediately leads to the desired result. Recall (a) that Sample expressions are countable because Samples consist of portions that form a partition over the material in the complement, and (b) that regular Pointers like *hint* do not guarantee that the corresponding portions are non-overlapping, preventing countability. If the abstract complement denotes a kind, each Pointer of a certain concept will be a manifestation of the same kind. In other words, the entity that is manifested is the same for each Pointer – therefore, the manifested items are overlapping, again preventing countability.

For the high reading, note that there does not appear to be a truth-conditional distinction between a *feeling of helplessness* and *feelings of helplessness*. The distinction rather seems to be the degree to which the experience is felt to be multifaceted. This is reminiscent of plurals of abundance, such as the *sands of the Sahara*, or the use of semantic agreement with group nouns (*the committee have ...*; Corbett 2006: pp. 155–160): here, too, the singular is acceptable, and the plural is used to emphasize plurality without changing truth conditions.

This provides a parallel with nouns of the type of *house* and *project*, discussed in section 5. Whereas a *house* can be treated as a collection of rooms, a *feeling* can be described as a collection of feelings. What we have in mind is a semantic map, or a state space (Churchland 1986), of feelings, where similar feelings are closer to each other. A particular feeling (say, happiness) may then be described in terms of its adjacent feelings (e.g., satisfaction, pleasure, harmony; Kövecses 2008). In fig. 1a above, one cannot take out the outline of room a and be left with an outline of room b: the remainder $\{t, u, v\}$ is no outline at all. Similarly, taking out one feeling from a network of feelings does not leave us with a coherent set of feelings.

Therefore, we argue that world knowledge about feelings and related classifiers prevents countability: specifically, the world knowledge that one feeling overlaps with its neighbors in a semantic map of emotional states. There is a crucial difference from *house* though: whereas *house* occurs as a complement in *the outlines of a house*, *feeling* occurs as a classifier in *a feeling of helplessness*. As a result, world knowledge about feelings prevents countability in the high reading, and not the low reading.

Combined with the fact that the complement is kind-denoting and the low reading therefore never provides a counting base without overlap, this explains why these classifier expressions never occur with numerals, in neither the high nor the low reading.

7 Related work: Metaphorical Measure Expressions

In this paper we have provided an analysis of a class of nominal expressions that can be modified by the indefinite article, can be pluralized, but cannot be modified by numerals. The same incomplete counting pattern has been discussed by Sutton et al. (2022) in relation to "metaphorical measure expressions":

(18) *The FBI has a heap/heaps of information about Sam.* (Sutton et al. 2022: p. 827)

(19) a. *The FBI and NSA each have a heap of information about Sam.* (Sutton et al. 2022: p. 827)
b. *# The US government has two heaps of information about Sam.*

Sutton et al. treat *heap of information* as a special kind of measure expression, the denotation of which lacks the set-theoretical property that allows for modification by numerals. Whereas regular measure expressions involve an equality with a relatively precise value (e.g., $\mu_{\mathrm{kg}}(\mathrm{x}, \mathrm{P}) = 1$ for *one kilo*), metaphorical measure expressions always refer to large or small amounts, and thus involve an inequality (i.e., $\mu(\mathrm{x}, \mathrm{P}) < \mathrm{n}$ or $\mu(\mathrm{x}, \mathrm{P}) > \mathrm{n}$, where n is a context-dependent threshold). The set of amounts of material (e.g., *information*) that satisfy such an inequality contains overlapping elements, and therefore does not provide a counting base with the properties required for modification by numerals.

There are some syntactic differences that justify keeping metaphorical measure expressions as separate from the classifier expressions we have discussed here. First, Sutton et al. (2022: p. 826) note that metaphorical measure expressions only occur with mass nouns and bare plurals as their complement (*a heap of* (**the*) *information*). By contrast, classifiers accept full-fledged DPs as well:[12]

(20) a. *a sense of* {*a life* / *helplessness*}
b. *the outline*(*s*) *of a project*
c. (*an*) *outline of legislation* / *psychoanalysis*

Furthermore, Sutton et al. (2022: pp. 827–878) observe that metaphorical measure expressions can be intensified, using a conjunction for large quantities (*heaps and heaps of information*) or a pseudo-partitive for small quantities (*an iota of an iota of information*). These constructions are unavailable with classifier expressions (**a drop of a drop of water*; **feelings and feelings of helplessness*).

Thus, Sutton et al.'s analysis complements our own. Like Chierchia's (2010) analysis of *quantity of*, it corresponds to our low counting base reading, as the counting base consists of portions of material. For Sutton et al., a *heap of information* is nothing more than a large quantity of information. However, we are concerned here with cases where nouns like *heap* have a classifying function in addition to their quantifying function, and thus also refer to shape and other spatial properties. These require us to introduce separate count entities that stand in a consists-of or manifests relation to a portion. Metaphorical measure expressions can be seen as bleached classifier expressions, where this classifying function and the corresponding count entity has been lost. As a result, only the low counting base reading remains.

8 Looking further afield

So far, we have concerned ourselves with the countability properties of *Cl of N* constructions, which we argue can be explained from the presence or absence of (different kinds of) overlap. In this section we show, without working out the details, that our notions of manifestation, overlap, and (structured) internal plurality have wider applicability.

8.1 Morphological reflexes of overlap

First, the kinds of internal plurality we have discussed have parallels in PLURALIA TANTUM (cf. Jouitteau & Rezac 2015: p. 576). Received wisdom holds that words like *glasses*, *pants*, and *scissors* are morphosyntactically plural because they consist of two salient parts. Yet we cannot talk about, say, #*two glasses*, to refer to those parts. This is similar to the way we can talk about *outlines* which are internally complex, even though the parts cannot be counted: the unavailability of the low reading can be explained from the fact that the parts are complementarily connected. In other words, connectedness can be seen as a type of minimal overlap between two entities, as it involves a part that connects the two and thus belongs to both (e.g., the wall connecting two rooms in the outlines of a house).

Other types of concepts that frequently surface as pluralia tantum (Koptjevskaja-Tamm & Wälchli 2001: p. 631) also denote internally complex entities for which the low reading is unavailable, for similar reasons. One cannot count **twenty Cotswolds*, because one cannot draw boundaries around the individual *wold-s*, just as one cannot draw boundaries around overlapping *feelings of helplessness*, and we cannot even count *the* (#*fifty*) *United States of America* (rather: *the fifty united states making up the USA*), because,

[12] Some nouns used in metaphorical measure expressions seem to accept full-fledged DPs as their arguments: *a heap of a task*. However, in such a case the complement is not divided into portions, but is qualified by the head.

as with *outlines* and *contours*, the parts are not independent but form a complex whole with its own laws and government. The parts of periods of time and festivities (German *Weihnachten* 'Christmas') cannot be counted because they stand in a fixed order, which provides structure; similarly, French *eaux* 'waters' in *prendre les eaux* 'follow a spa treatment' (lit. 'take the waters') refers to a standardized sequence of treatments involving water (and here, too, we cannot talk about the **cinq eaux de Vichy* 'five waters of Vichy'). Some diseases are thought to receive plural morphology because they present themselves in multiple places or via multiple symptoms (e.g., *measles*, *mumps*, *rickets*, *shingles*). These symptoms are essentially Pointers to the abstract disease; overlap in the manifested item (the kind *measles*) thus prevents countability (we owe this observation to Marcin Wągiel). Finally, the multiple actions or participants that explain plural morphology for certain activities (*billiards*, *checkers*) are not independent but involve rule-based turn-taking: this provides the internal plurality with a structure akin to that of *outlines of a house*, making it uncountable.

Languages struggle with expressing counted pluralia tantum. For example, *glasses* cannot be counted without the insertion of the classifier *pair*: *two* #(*pairs of*) *glasses*. It is not exactly clear to us what excludes the high reading in *two glasses* – that is, why *two glasses* cannot refer to two pairs of glasses. However, many languages require special syntax to count pluralia tantum (Corbett 2019: pp. 92–94; and see Allan 1980: p. 557 for combination with quantifiers).

Second, in some languages, overlap may be implied by the choice of one of several PLURAL ALLOMORPHS. Dutch nouns have a plural on either *-s* or *-en*, which is typically selected based on phonological rules. Some nouns accept both plurals, and for a subset of these, the two forms are semantically distinct. In these cases, the *-en* plural refers to an internally complex plural, and does not combine with numerals. Thus, the *-en* plural *wapenen* 'weapons' is used in idioms like *onder de wapenen* 'in military service' (lit. 'under the weapons') and *de wapenen opnemen* 'take up arms'. It only refers to complementary types of weapons, and cannot refer to a set of identical weapons. Only *wapens* can be used to refer to a homogeneous (non-complementary) collection of identical weapons, and only *wapens* can be modified by numerals (regardless of whether the weapons in the denoted set are of the same or of a different type). The complementariness in *wapenen* provides a structure similar to the way in which rooms complement each other in a house (see section 5), explaining the incompatibility of numerals. Similar considerations extend to *hemelen* 'heavens' and (*internationale*) *wateren* 'international waters'. Profession nouns on *-aar* generally have both plurals, but the *-en* plural is degraded for professions in which people do not typically collaborate. Thus we have *dienaren* 'servants' but not ?*molenaren* 'millers' or ?*moordenaren* 'murderers': collaboration is needed to provide the complementary structure reflected in the *-en* plural. The size of the group of nouns sensitive to this kind of internal plurality remains unclear at this point, as does the strength of the internal complexity implicature of *-en*.

8.2 Parallels in the verbal domain

It is well-known that the mass/count distinction has similarities with the atelic/telic distinction in the verbal domain (Krifka 1989, Champollion 2015 and references therein). A natural question to ask is therefore what the notion of overlap and structure would look like in the domain of events. We make three suggestions here, the first two due to Marcin Wągiel.

First, MANIFESTING VERBS like *reflect* or *mirror* are reminiscent of Pointers like *hint* (section 4) in that they manifest something without any effect on the thing manifested. Not accidentally, these verbs come from the same subjective sensory domain as Pointers like hint. The behavior of these verbs that is of interest here is that (being stative) they reject frequentative adverbials like *twice*, but do accept adverbials like *in two ways*:

(21) a. *These actions reflected her generosity* *twice / *in two ways.*
b. *His living room mirrored his personality* *twice / *in two ways.*

This can be said to be parallel to the acceptance of numerals with Pointers like *hint* when different types of hints (i.e., different ways for a material to manifest itself) are distinguished, such as an olfactory and a gustatory hint of cognac.

The second parallel comes from PLURACTIONAL VERBS. Observe first that verbs like *knock* are ambiguous between a semelfactive and an iterative reading. In the latter, they denote internally complex events: *Sue knocked on the door* may be used to describe an event in which Sue knocked two or three times immediately after each other, as long as these knocks are seen as constituting a single knocking event. This is somewhat similar to a *feeling*, which can be described in terms of other feelings (section 6), and to *outline*, of which the plural implies greater internal elaboration (section 5). Therefore, when such verbs combine with frequentative adverbials like *twice*, there are in principle two readings possible: in the low reading, the adverbial targets the internal plurality (the number of knocks in the iterative event); in the high reading, it targets the external plurality (separate knocking events which may each be internally complex). In English, *Sue knocked two times* is ambiguous: it can describe a situation in which Sue drew attention once by making two sounds close together (the low reading), but also a situation in which Sue knocked once, got no reply, and knocked again (the high reading).

By contrast, there is also a class of pluractional verbs that only seem to permit the high reading. When a loudspeaker *crackles*, it makes a sequence of crackling sounds. The sounds that make up a crackle cannot be individuated: they overlap. For this reason, in *the loudspeaker crackled twice and went silent*, there must be silence in between two crackling sequences: only the high reading is available; overlap prevents the low reading. Other verbs with this property are *flicker*, *glitter*, *murmur*, *nibble*, and *rattle*. However, in *the engine sputtered twice before shutting down*, the adverbial does seem to be able to target the internal plurality, referring to two short "sput"s: these sounds have a clearer boundary, and therefore overlap less, than those constituting a *crackle*.[13]

Third, there are also parallels with languages with numeral classifiers. Ye & Guo (2023) discuss a classifier construction in Mandarin Chinese for event counting, which has a different interpretation depending on the modified verb. With verbs that denote stably atomic events, such as *qiao* 'knock', the construction *qiao yi xia* 'knock one CL' has a pluractional interpretation: 'give some knocks'. By contrast, with verbs that are not stably atomic, this interpretation is unavailable, and the construction instead gives rise to a durative reading: *deng yi xia* 'wait a while' (rather than 'wait several times'). The latter is comparable to the way *outlines of a house* does not refer to several houses and instead makes the fact that the house consists of several parts salient. In the context of the head noun *outline*, a *house* can be seen as not stably atomic, as it divides into parts that each also have an outline. Similarly, durative events can be seen as consisting of non-atomic intervals.

In sum, it is clear that the notions of both manifestation and internal plurality have parallels in other domains.

9 Conclusion

In this paper we have discussed four different kinds of classifier expressions. These can be categorized as Samples and three kinds of Pointers: those with mass/bare plural complements (section 4), those with (structured) DP complements (section 5), and those with abstract complements (section 6).

Samples, like *drop of water*, denote entities that consist of portions of material. These portions form a partition: the portions corresponding to two distinct Samples do not overlap. Pointers lack this property: they denote entities that are manifestations of portions, and two Pointers may be manifestations of the same portion, or of overlapping portions. This difference explains the main difference in countability

[13]In their survey of verbal diminutives, Audring et al. (2021: pp. 244–245) mention several languages with similar morphologically marked iterative verbs, which thus distinguish between external (high) and internal (low) plurality (Cusic 1981). We do not know to what extent the availability of high and low readings of frequentative adverbials generalizes over these languages, although the French data (with e.g. *taper* 'tap once/several times' – *tapoter* 'tap several times') are comparable to what we see in English.

	Samples	Pointers with… mass/bare plural	structured DP	abstract noun
low counting base	✓ partition	✗ overlapping portions	✗ structure	✗ identical kind
high counting base	✓ atoms	✓ atoms	✓ atoms	✗ structure

Table 1: Availability of modification by numerals for different constructions and readings.

between the two, in default contexts: we can count Samples (*two drops of water*) but not Pointers (#*two hints of cognac*), because only the former have a counting base with mutually disjoint members.

Most Pointer expressions can be modified by numerals in specific contexts. This depends on the existence of a separate counting base. For Pointers with mass/bare plural complements (*hints of cognac*), the counting base is forced to be in the domain of the Pointer entity (rather than that of the material it manifests). Thus, we can count different kinds of Pointers (e.g. gustatory and olfactory *hints*). Or we can count the events in which these Pointers are experienced.

Pointers with DP complements occur with specific complements that have a certain inner structure, such as a *house* (which consists of adjacent rooms) or a *project* (which consists of related subprojects). The classifier (*outline*, *contour*) is such that it can target both the complement itself and its parts. As a result, pluralizing the classifier triggers a reinterpretation of the complement as "internally plural": *the outlines of a house* refers to a detailed floor plan – since this kind of plurality involves overlap, it does not allow for modification by numerals. With a high counting base, we can still talk of *two outlines of a house*, which are drawn separately.

Finally, Pointers with abstract complements (*a feeling of helplessness*) necessarily manifest overlapping entities, since they are all manifestations of the same kind denoted by the abstract noun (rather than of a portion). Furthermore, for these classifiers, the high reading does not provide a suitable counting base either: we suggest that this is because these classifiers have a certain recursive structure: just like a house consists of rooms, a feeling "consists of" (rather: can be described in terms of) other feelings that are not independent of each other. Thus, the argument that prevents countability in the low reading for *outlines of a house* prevents countability in the high reading for *feelings of helplessness*.

Our analysis of the low reading of Pointer expressions (and the high reading of Pointers with abstract complements) depends on a new condition on the use of numerals, stated formally in (8b-iii), repeated here. In natural language, this says that there may not be any overlap between the elements of the set that are counted.

(8b-iii) the P-atoms in x do not overlap: $\forall a, b \in x \cap P[a \neq b \rightarrow a \cap b = \varnothing]$

Overlap may arise in different ways. The availability of countability in the high and low reading for each construction is summarized in table 1, where we also indicate the type of overlap preventing countability in each construction.

It appears that the low counting base reading, which is based on the material in the complement, is the default in neutral contexts. This can be seen as reflecting the primacy of space in the human mind.

Although our analysis is largely based on English, we expect that the following pattern from table 1 generalizes to other languages:

(22) Numerals can target internal plurality (the low reading) only if they can also target external plurality (the high reading) with the same complement in a different context.

However, languages may differ as to the types of overlap that do/don't provide for a suitable counting base. For instance, Italian morphosyntax also seems to be sensitive to the property of structure. A small class of masculine nouns has, besides the regular masculine plural, an irregular feminine plural

(Acquaviva 2008; Wągiel 2021: pp. 46–57). The feminine form carries an implicature that the entities form a collective, e.g. *mura* 'walls (in a connected complex)' (vs. masculine *muri* 'walls'). However, this structure does not have an effect on countability. Both plurals can be modified by numerals, although the distribution is not equal (e.g., four walls are typically a collective, marked by *mura*, whereas two walls are more often referred to by *muri*).

We hope that the analysis of the countability patterns of different types of classifier expressions presented here has shed some light on the ontological properties that play a role in quantification.

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